

5(a)	any three from: 1 cDNA has, no introns / only exons or genomic DNA has introns (and exons) ; 2 introns are non-coding sequences / exons are coding sequences ; 3 bacteria cannot, remove introns / carry out splicing ; 4 with cDNA, functional mRNA is made / translation occurs or with genomic DNA, functional mRNA is not made / translation does not occur ;	3
5(b)(i)	bond together / anneal, with hydrogen bonds ; formation of phosphodiester bonds / joins sugar phosphate backbone ;	2
5(b)(ii)	1 so RNA polymerase can bind ; 2 so, transcription / gene expression, occurs ;	2

Question	Answer	Marks
5(c)	any three from: 1 marker gene is added to plasmid ; 2 positioned downstream of promoter ; 3 positioned alongside, GOI / insulin gene; 4 when marker gene is expressed fluorescent proteins are made ; 5 <i>ref. to</i> (fluorescent proteins) fluoresce / glow, with exposure to, UV / blue, light ; 6 shows which, bacteria / cells, are transformed / AW ;	3
5(d)	any two from: <i>advantages of using yeast:</i> 1 easier / cheaper, to culture yeast cells ; 2 more productive / higher productivity ; 3 easier, to extract / processing / to purify, (insulin) ; 4 yeast cell similar to β-cells / named example ; e.g. have Golgi apparatus	2

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